Artificial intelligence

Artificial intelligence (**AI**) is the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making. It is a field of research in engineering, mathematics and computer science that develops and studies methods and software that enable machines to perceive their environment and use learning and intelligence to take actions that maximize their chances of achieving defined goals.[1]

High-profile applications of artificial intelligence include advanced web search engines (e.g., Google Search); recommendation systems (used by YouTube, Amazon, and Netflix); virtual assistants (e.g., Google Assistant, Siri, and Alexa); autonomous vehicles (e.g., Waymo); generative and creative tools (e.g., language models and AI art); and superhuman play and analysis in strategy games (e.g., chess and Go).

The traditional goals of AI research include learning, reasoning, knowledge representation, planning, natural language processing, and perception, as well as support for robotics.[a] To reach these goals, AI researchers have used techniques including state space search and mathematical optimization, formal logic, artificial neural networks, and methods based on statistics, operations research, and economics.[b] AI also draws upon psychology, linguistics, philosophy, neuroscience, and other fields.[2] Some companies, such as OpenAI, Google DeepMind and Meta, aim to create artificial general intelligence (AGI) – AI that can complete virtually any cognitive task at least as well as a human.[3]

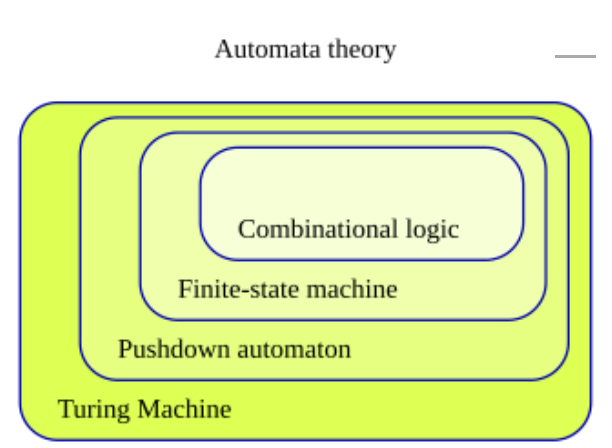
Artificial intelligence was founded as an academic discipline in 1956,[4] and the field went through multiple cycles of optimism throughout its history,[5][6] followed by periods of disappointment and loss of funding, known as AI winters.[7][8] Funding and interest increased substantially after 2012, when graphics processing units began being used to accelerate neural networks, and deep learning outperformed previous AI techniques.[9] This growth accelerated further after 2017 with the transformer architecture.[10] In the 2020s, an AI boom has coincided with advances in generative AI, which allowed for the creation and modification of media. In addition to AI safety and unintended consequences and harms from the use of AI, ethical concerns, AI's long-term effects, and potential existential risks have prompted discussions of AI regulation.

Goals

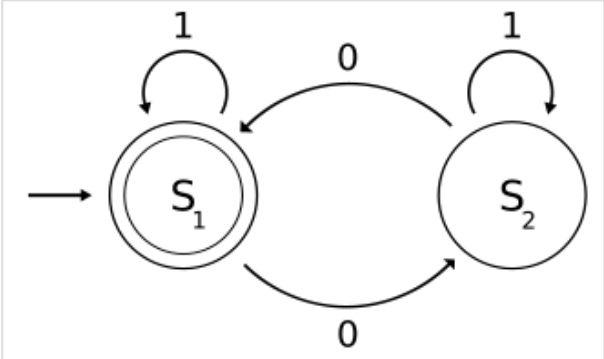
The general problem of simulating (or creating) intelligence has been broken into subproblems. These consist of particular traits or capabilities that researchers expect an intelligent system to display. The traits described below have received the most attention and cover the scope of AI research.[a]

Automata theory

Automata theory is the study of abstract machines and automata, as well as the computational problems that can be solved using them. It is a theory in theoretical computer science with close connections to cognitive science and mathematical logic. The word *automata* comes from the Greek word αὐτόματος, which means "self-acting, self-willed, self-moving". An automaton (automata in plural) is an abstract self-propelled computing device which follows a predetermined sequence of operations automatically. An automaton with a finite number of states is called a finite automaton (FA) or finite-state machine (FSM). The figure on the right illustrates a finite-state machine, which is a well-known type of automaton. This automaton consists of states (represented in the figure by circles) and transitions (represented by arrows). As the automaton sees a symbol of input, it makes a transition (or jump) to another state, according to its transition function, which takes the previous state and current input symbol as its arguments.



Classes of automata



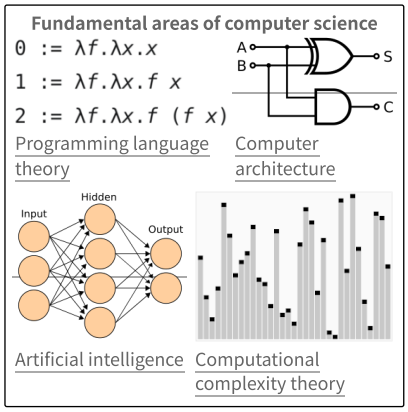
The automaton described by this state diagram starts in state S_1, and changes states following the arrows marked 0 or 1 according to the input symbols as they arrive. The double circle marks S_1 as an accepting state. Since all paths from S_1 to itself contain an even number of arrows marked 0, this automaton accepts strings containing even numbers of 0s.

Automata theory is closely related to formal language theory. In this context, automata are used as finite representations of formal languages that may be infinite. Automata are often classified by the class of formal languages they can recognize, as in the Chomsky hierarchy, which describes a nesting relationship between major classes of automata. Automata play a major role in the theory of computation, compiler construction, artificial intelligence, parsing and formal verification.

History

The theory of abstract automata was developed in the mid-20th century in connection with finite automata.[1] Automata theory was initially considered a branch of mathematical systems theory, studying the behavior of discrete-parameter systems. Early work in automata theory differed from previous work on systems by using abstract algebra to describe information systems rather than differential calculus to describe material systems.[2] The theory of the finite-state transducer was

Computer science



Computer science is the study of computation, information, and automation.[1][2][3] Included broadly in the sciences, computer science spans theoretical disciplines (such as algorithms, theory of computation, and information theory) to applied disciplines (including the design and implementation of hardware and software).[4][5][6] An expert in the field is known as a computer scientist.

Algorithms and data structures are central to computer science.[7] The theory of computation concerns abstract models of computation and general classes of problems that can be solved using them. The fields of cryptography and computer security involve studying the means for secure communication and preventing security vulnerabilities. Computer graphics and computational geometry address the generation of images. Programming language theory considers different ways to describe computational processes, and database theory concerns the management of repositories of data. Human–computer interaction investigates the interfaces through which humans and computers interact, and software engineering focuses on the design and principles behind developing software. Areas such as operating systems, networks and embedded systems investigate the principles and design behind complex systems. Computer architecture describes the construction of computer components and computer-operated equipment. Artificial intelligence and machine learning aim to synthesize goal-orientated processes such as problem-solving, decision-making, environmental adaptation, planning and learning found in humans and animals. Within artificial intelligence, computer vision aims to understand and process image and video data, while natural language processing aims to understand and process textual and linguistic data.

The fundamental concern of computer science is determining what can and cannot be automated.[2][8][3][9][10] The Turing Award is generally recognized as the highest distinction in computer science.[11][12]

History

The earliest foundations of what would become computer science predate the invention of the modern digital computer. Machines for calculating fixed numerical tasks such as the abacus have existed since antiquity, aiding in computations such as multiplication and division. Algorithms for performing computations have existed since antiquity, even before the development of sophisticated computing equipment.[16]

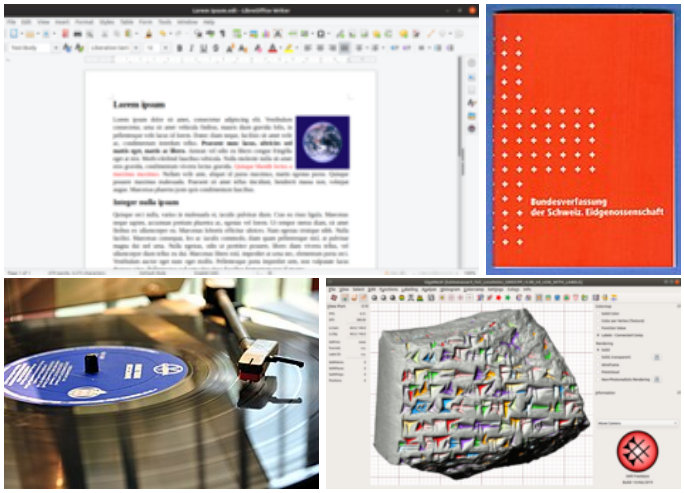
Gottfried Wilhelm Leibniz (1646–1716) developed logic in a binary number system and has been called the "founder of computer science".[13]

Wilhelm Schickard designed and constructed the first working mechanical calculator in 1623.[17] In 1673, Gottfried Leibniz demonstrated a digital mechanical calculator, called the Stepped Reckoner.[18][19] Leibniz may be considered the first computer scientist and information theorist, because of various reasons, including the fact that he documented the binary number system. In 1820, Thomas de Colmar launched the mechanical calculator industry[note 1] when he invented his simplified arithmometer, the first calculating machine strong enough and reliable enough to be used daily in an office environment. Charles Babbage started the design of the first *automatic mechanical calculator*, his Difference Engine, in 1822, which eventually gave him the idea of the first *programmable mechanical calculator*, his Analytical Engine.[20] He started developing this machine in 1834, and "in less than two years, he had sketched out many of the salient features of the modern computer".[21] "A crucial step was the adoption of a punched card system derived from the Jacquard loom"[21] making it infinitely programmable.[note 2] In 1843, during the translation of a French article on the Analytical Engine, Ada Lovelace wrote, in one of the many notes she included, an algorithm to compute the Bernoulli numbers, which is considered to be the first published algorithm ever specifically tailored for implementation on a computer.[22] Around 1885, Herman Hollerith invented the tabulator, which used punched cards to process statistical information; eventually his company became part of IBM. Following Babbage, although unaware of his earlier work, Percy Ludgate in 1909 published[23] the 2nd of the only two designs for mechanical analytical engines in history. In 1914, the Spanish engineer Leonardo Torres Quevedo published his *Essays on Automatics*,[24] and designed, inspired by Babbage, a theoretical electromechanical calculating machine which was to be controlled by a read-only program. The paper also introduced the idea of floating-point arithmetic.[25][26] In 1920, to celebrate the 100th anniversary of the invention of the arithmometer, Torres presented in Paris the Electromechanical Arithmometer, a prototype that demonstrated the feasibility of an electromechanical analytical engine,[27] on which commands could be typed and the results printed automatically.[28] In 1937, one hundred years after Babbage's impossible dream, Howard Aiken convinced IBM, which was making all kinds of punched card equipment and was also in the calculator business[29] to develop his giant programmable calculator, the ASCC/Harvard Mark I, based on Babbage's Analytical Engine, which itself used punched cards and a central processing unit. When the machine was finished, some hailed it as "Babbage's dream come true".[30]

Charles Babbage is sometimes referred to as the "father of computing".[14]

During the 1940s, with the development of new and more powerful computing machines such as the Atanasoff–Berry computer and ENIAC, the term *computer* came to refer to the machines rather than their human predecessors.[31] As it became clear that computers could be used for more than just mathematical calculations, the field of computer science broadened to study computation in general. In 1945, IBM founded the Watson Scientific Computing Laboratory at Columbia University in New York City. The renovated fraternity house on Manhattan's West Side was IBM's first laboratory devoted to pure science. The lab is the forerunner of IBM's Research Division, which today operates research facilities around the world.[32] Ultimately, the close relationship between IBM and Columbia University was instrumental in the emergence of a new scientific discipline, with Columbia offering one of the first academic-credit courses in computer science in 1946.[33] Computer science began to be established as a distinct academic discipline in the 1950s and early 1960s.[34][35] The world's first computer science degree program, the Cambridge Diploma in Computer Science, began at the University of Cambridge Computer Laboratory in 1953. The first computer science department in the United States was formed at Purdue University in 1962.[36] Since practical computers became available, many applications of computing have become distinct areas of study in their own rights.

Document



Documents across mediums. **Top-left**: a word processor document using LibreOffice. **Top-right**: a copy of the Swiss Constitution in German. **Bottom-left**: a vinyl record holding a set of songs. **Bottom-right**: a computer program interpreting a fragment of a clay tablet with cuneiform script about king Shalmaneser III

A **document** is a written, drawn, presented, or memorialized representation of thought, often the manifestation of non-fictional, as well as fictional, content.[1][2] The etymology of the word "document" derives from the Latin *documentum*, which denotes a "teaching" or "lesson": the verb *doceō* denotes "to teach". Historically, the term "document" was usually used to indicate written proof useful as evidence of a truth or fact.

In the Computer Age, the term "document" typically refers to a primarily textual computer file, encompassing its structural and format elements, such as fonts, colors, and images. In the contemporary era, the definition of "document" has expanded beyond its traditional medium, such as paper, to encompass electronic documents as well. History, events, examples, opinions, stories, and creativity can all be expressed in documents.

"Documentation" is distinct because it has more denotations than "document". Documents are also distinguished from "realia", which are three-dimensional objects that would otherwise satisfy the definition of "document" because they memorialize or represent thought. Documents are usually considered to be two-dimensional representations.

Abstract definitions

The concept of "document" has been defined by Suzanne Briet as "any concrete or symbolic indication, preserved or recorded, for reconstructing or for proving a phenomenon, whether physical or mental."[3]

An often-cited article concludes that "the evolving notion of *document*" among Jonathan Priest, Paul Otlet, Briet, Walter Schürmeyer, and the other documentalists increasingly emphasized whatever functioned as a document rather than traditional physical forms of documents. The shift

Parsing

Parsing, **syntax analysis**, or **syntactic analysis** is a process of analyzing a string of symbols, either in natural language, computer languages or data structures, conforming to the rules of a formal grammar by breaking it into parts. The term *parsing* comes from Latin *pars* (*orationis*), meaning part (of speech).[1]

The term has slightly different meanings in different branches of linguistics and computer science. Traditional sentence parsing is often performed as a method of understanding the exact meaning of a sentence or word, sometimes with the aid of devices such as sentence diagrams. It usually emphasizes the importance of grammatical divisions such as subject and predicate.

Within computational linguistics the term is used to refer to the formal analysis by a computer of a sentence or other string of words into its constituents, resulting in a parse tree showing their syntactic relation to each other, which may also contain semantic information. Some parsing algorithms generate a *parse forest* or list of parse trees from a string that is syntactically ambiguous.[2]

The term is also used in psycholinguistics when describing language comprehension. In this context, parsing refers to the way that human beings analyze a sentence or phrase (in spoken language or text) "in terms of grammatical constituents, identifying the parts of speech, syntactic relations, etc."[1] This term is especially common when discussing which linguistic cues help speakers interpret garden-path sentences.

Within computer science, the term is used in the analysis of computer languages, referring to the syntactic analysis of the input code into its component parts in order to facilitate the writing of compilers and interpreters. The term may also be used to describe a split or separation.

In data analysis, the term is often used to refer to a process extracting desired information from data, e.g., creating a time series signal from a XML document.

Human languages

Traditional methods

The traditional grammatical exercise of parsing, sometimes known as *clause analysis*, involves breaking down a text into its component parts of speech with an explanation of the form, function, and syntactic relationship of each part.[3] This is determined in large part from study of the language's conjugations and declensions, which can be quite intricate for heavily inflected languages. To parse a phrase such as "man bites dog" involves noting that the singular noun "man" is the subject of the sentence, the verb "bites" is the third person singular of the present tense of the verb "to bite", and the singular noun "dog" is the object of the sentence. Techniques such as sentence diagrams are sometimes used to indicate relation between elements in the sentence.